ROHAN KATKAM

MSc in Embedded Systems, TU Delft, The Netherlands
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EDUCATION

Delft University of Technology, The Netherlands (August 2020 - March 2023)
Master of Science, Embedded Systems
Specialization in Software and Networking track

National Institute of Technology Karnataka, India (August 2016 - June 2020)
Bachelor of Technology, Electronics and Communications Engineering

WORK EXPERIENCE

Software Developer, Rambus, The Netherlands (July 2024 - Present)

- Develop x86 assembly firmware for a Root of Trust (RoT) IP in SoC environments, implementing industry standard cryptographic algorithms
- Contribute to verification spec development, Python-based regression test frameworks for IP debugging, and validation
- Worked on integration of non-NIST schemes like PAKE and contributed to DDK development for additional IPs using C.

Embedded Software Engineer, Signify N.V., The Netherlands (March 2023 - March 2024)

- Developed embedded firmware in C for wireless sensors in lighting systems (EFR32, ARM Cortex-M, Micrium OS), communication involved Bluetooth, Zigbee, and DALI protocols
- Integrated a third party microcontroller diagnostics SDK for logging metrics, and fault detection.
- Contributed to agile workflows (Jira, Confluence), and Git versioning processes

Student Intern, Laevo B.V., The Netherlands (July 2021 - September 2021)

- Supported development of a machine learning classifier for a passive exoskeleton system. Prototyped pipelines for IMU data acquisition across Adruino/STM platforms
- Implemented WiFi and NB-IoT communication layers for cloud synchronization

Summer Intern, University Centre for Energy Efficient Buildings, Czech Technical University, Czechia (June 2019 - August 2019)

- Processed and compressed sky-imaging data using Python and OpenCV, improving storage efficiency by 2× while preserving critical information

PROJECTS

LC-Camera Based Visible Light Communication (VLC) (February 2022 - March 2023)
MSc Thesis, TU Delft

- Developed a multi-pixel multi-channel VLC based system to modulate liquid crystals (LC) under ambient light, and decoder at the camera receiver
- Improved the per-pixel the data throughput under various lighting, distance and viewing conditions

Removing sensitive data from panoramic imagery (March 2021 - April 2021)
Project in Deep Learning

- Successfully reproduced a project by the municipality of Amsterdam to remove sensitive information like faces and vehicle plates from high-res panoramic street images.

· Utilized object detection models like YOLOv5, Faster R-CNN, and pre-trained YOLOv5, and tested model predictions on similar images from Google Street View and self-taken images in Delft to evaluate generalization.

Testbed Development for Industrial Systems (August 2019 - December 2019)
Bachelor Internship and Thesis, Indian Institute of Science, India

· Created a mecanum-wheeled robot for navigating a mapped environment; employed kinematics and data fusion from IMUs, encoders and UWB sensors
· Developed a linear actuator delta robot and its construction of the inverse kinematic model
· Developed a software for a 3D printing system: Parsed development build files to equipment specific formats in Python; Created a front-end visualizer for the industrial process with PyQtGraph and OpenGL

RELEVANT COURSES

Masters: Embedded Software; Wireless IoT and Local Area Networks; Real-Time Systems; Smart Phone Sensing; Wireless Communication; Mobile Networks; Deep Learning, Robotics Software;
Bachelors: Digital Electronics; Computer Architecture; Microprocessors; Signal Processing; Digital System Design; Adhoc Networks;

TECHNICAL SKILLS

Software Tools:	LLMs, Version Control: MKS, Git, Atlassian tools, Jenkins, KiCAD
Programming Practices:	System Architecture Translation, Object Oriented Programming, CI/CD, Agile, Real-time OS
Programming Languages:	C, C++, x86 Assembly, Python, Bash
Embedded Platforms:	EFR32, STM32, Atmega, Raspberry Pi, ESP8266, Xilinx Artix-7 FPGA

ACHIEVEMENTS

1) Finalists, Nationals Round, Present Around the World (PATW) Competition (2018)
2) Special Mentions, NITK Model United Nations (2017)
3) Gold Medalist, Mathematics, National Talent Search Examination (2014)

OTHER ACTIVITIES

BuurtBoerderij De Nijkamphoeve, Den Haag (2022)

· Dedicated to animal care in a farm on a voluntary basis during the weekends.

Institution of Engineering and Technology (2017 - 2020)
Publicity Coordinator, Executive Member

· Lead the electronics group, engaged in hosting technical workshops, and student based projects.